A Report File

On

Active Directory Users And Computers



FOUR MONTHS INDUSTRIAL TRAINING REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

(COMPUTER SCIENCE & ENGINEERING)

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2024

DECALARATION

I hereby certify that the work which is being presented in the project entitled "**ACTIVE DIRECTORY USERS AND COMPUTERS**" by **EKAMPREET SINGH** in partial fulfillment of requirements for the award of degree of **B.Tech** submitted in the department of **Computer Science and Engineering** at University Institute Of Engineering and Technology under **Sant Baba Bhag Singh University, Jalandhar** is an authentic record of my own work carried out during a period from June to July 2026, under the supervision of MR.KHUSHDEEP SINGH. The matter presented in this project has not been submitted by me in any other University / Institute for the award of B.Tech degree.

Signature of the student

This is to certify that the above statement made by the candidate is correct to the best of my / our knowledge.

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ABSTRACT

The Active Directory Users and Computers (ADUC) project provides a centralized, secure, and user-friendly solution designed to simplify and optimize the management of network identities, user accounts, and IT resources within an organization. By replacing fragmented user management with a structured directory service, the project enables IT teams to maintain tight control over system access while ensuring smooth daily operations. The system is structured into three primary operational tiers—Directory Administration, Organizational Unit (OU) & Role Management, and User Access & Policy Governance—each tailored to meet specific administrative needs and keep network workflows running seamlessly.

The Directory Administration module serves as the backbone of the systemWith a single control center, administrators maintain absolute control over the central database, establishing a secure and reliable framework for identity verification across the entire corporate network.

The Organizational Unit (OU) & Role Management module equips IT staff and department managers with essential tools to maintain order within the network hierarchy. Through this interface, staff can group users and computers into specific OUs based on departments, grant delegation rights, and assign security groups. This structured approach reduces administrative overhead, minimizes human error, and makes IT housekeeping significantly easier.

The User Access & Policy Governance layer focuses on delivering a seamless yet protected experience for end-users and endpoints. By enforcing Group Policy Objects (GPOs), the system manages password policies, software distribution, and resource permissions across all connected computers. Upon logging into their workstation, users gain appropriate, secure access to shared drives, network printers, and applications tailored to their specific job roles.

Built using modern security standards and industry best practices, the system emphasizes scalability, robust authentication, and strict access controls. By integrating these key identity management tools into a cohesive platform, the project significantly reduces manual IT errors, enhances organizational security, and fosters a structured approach to domain administration. Ultimately, this project highlights how modern directory services can transform enterprise IT infrastructure and elevate security for all stakeholders.

About the Company/Industry

Sensation Software Solutions Pvt. Ltd.

Do Elegant Engineering | An IT Company Since 2013

Sensation Software Solutions Pvt. Ltd. is a established IT company and technology training provider committed to delivering high-performance, secure, and scalable solutions for modern enterprises worldwide. Operating at the forefront of digital engineering, we specialize in building enterprise-grade software, advancing cloud infrastructure, and training professionals across key tech disciplines—including Cyber Security, AI & Machine Learning, Data Science, and Full Stack Development. Our focus is on centralizing identity access management, enforcing strict network controls, and engineering robust Active Directory architectures that safeguard corporate assets and drive operational efficiency.

Our mission is to empower businesses and aspiring professionals through transformative digital solutions and industry-aligned training that foster innovation, elevate security, and deliver tangible value. We prioritize excellence, practical execution, and security above all else—ensuring every project and training program meets the highest standards of quality.

We offer a comprehensive range of IT services and professional domain specializations designed to address complex modern technology challenges:

- Cyber Security & Active Directory (AD) Management
- Full Stack Development & Web/Graphic Designing
- AI, Machine Learning & Data Science
- Cloud Computing & Infrastructure Solutions
- Quality Assurance, Data Analytics & Business Analysis
- Software Maintenance, System Auditing & Managed IT Services

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Chapter-1

Introduction To Organization

1.1 Industry Profile

Overview of Cyber Security Industry

The global Information Technology (IT) and Cyber Security industry represents one of the most critical and rapidly expanding sectors in the modern digital economy. As organizations worldwide accelerate digital transformation, transition to hybrid working models, and migrate core services to cloud environments, securing network infrastructure, identity domains, and sensitive enterprise data has shifted from an operational concern to a strategic business imperative.

Cyber Security encompasses the practices, technologies, and processes designed to protect systems, networks, applications, and data from unauthorized access, cyberattacks, and data breaches. Within this broad industry, Identity and Access Management (IAM) and Directory Services (such as Microsoft Active Directory) serve as the fundamental security framework. They ensure that only authenticated and authorized users gain access to designated network resources and organizational computers.

❖ Market Size and Growth Trends-

- Global Market Growth: The global cybersecurity market size was valued at approximately USD 234.2 Billion and is projected to expand significantly over the coming years at a Compound Annual Growth Rate (CAGR) exceeding 11%.
- Indian Market Expansion: India has emerged as one of the fastest-growing cybersecurity markets globally. The Indian cybersecurity industry market size reached over USD 6.5 Billion, driven by nationwide initiatives such as *Digital India*, rapid adoption of online financial systems
- ms, strict government regulations, and increased cyber threat visibility across corporate networks.
- Dominance of Identity Access Management (IAM): Solutions focusing on identity governance, user privilege management, and directory access control represent one of the largest and fastest-growing software segments in the security market.

1.2 General Information

Overview of the Organization

Sensation Software Solutions Pvt. Ltd. *"Do Elegant Engineering"* is a professional IT services, custom software development, and specialized technology training company established in **2013**. The company provides end-to-end digital engineering solutions, ranging from web and mobile application development to enterprise-level cyber security and identity management services.

In addition to software development, the training division of the company offers industry-oriented training programs to engineering students in core technology areas such as Cyber Security, Active Directory Administration, Full Stack Development, Cloud Computing, Data Science, and Artificial Intelligence.

Key Operational Details

Parameter	Company Details
Full Company Name	Sensation Software Solutions Private Limited
Company Tagline	*Do Elegant Engineering*
Year of Establishment	2013
Key Leadership / Management	Mr. Rahul Adya (Director / CEO) Mr. Sudhir Singh (Director / CTO)
Core Business Domains	• Cyber Security & Identity Access Management (IAM) • Web & Mobile Application Engineering • IT Training, Internship & Skill Development
Registered Address	Office No. 26, Prosperity Square, Plot No. D-185, Phase 8-B, Industrial Area, Mohali (S.A.S. Nagar), Punjab - 160055
Official Email ID	sensationsolutions@gmail.com
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Chapter-2

Introduction To Project

2.1Need of Project

In modern enterprise companies and organizations, managing thousands of user accounts, staff permissions, and office computers is a major challenge for IT departments. Traditional or manual ways of managing users—like setting up password settings on every single computer individually or maintaining user lists on simple spreadsheets—are extremely inefficient, risky, and time-consuming.

Without a central control system, IT teams face several major issues:

- **Security Risks:** It is difficult to track who has access to sensitive files, leading to unauthorized access and data leaks.
- **Repetitive Manual Work:** If a new employee joins or changes departments, IT staff have to manually change settings on multiple individual systems.
- **Weak Password Control:** Users often set weak passwords because there is no automated rule or policy enforcing strong passwords across all office computers.
- **Lack of Central Control:** Disabling access for a resigned employee takes too much time, leaving system access open even after they leave the organization.

The **Active Directory Users and Computers (ADUC)** project was created to solve these exact problems. It provides a centralized digital control platform that allows IT administrators to manage all users, passwords, computers, and system permissions from a single dashboard. By automating user access and group policies, the system makes network management fast, error-free, and highly secure.

2.2 Objectives of the Project

This project is designed to achieve the following clear objectives:

1. **Establish Centralized Control:** Create a single, secure domain network where all user accounts, computers, and office resources are controlled from one central database.
2. **Automate User & Resource Management:** Streamline daily IT tasks, such as creating new user accounts, resetting forgotten passwords, and grouping computers into specific departments.

3. **Implement Role-Based Access Control (RBAC):** Provide appropriate permissions based on user job roles:
    - **Domain Administrators:** Have full control over the entire network, user accounts, and security settings.
    - **IT / Department Staff:** Can manage specific sub-groups (Organizational Units) and assist users with basic operational tasks.
    - **Standard End-Users (Employees):** Get secure access only to the specific files, shared folders, and applications needed for their work.
4. **Enforce Group Policy Objects (GPOs):** Automatically apply security rules (like strong password policies, screen-lock timers, and software restrictions) across all connected office PCs.
5. **Enhance System Transparency & Security:** Reduce manual errors, prevent unauthorized logins, and ensure real-time tracking of network activity across the entire organization.

2.3 Scope of the Project

The scope of the **Active Directory Users and Computers (ADUC)** project covers the design, configuration, and implementation of a centralized network management system for enterprise IT environments. The primary areas covered within this project include:

- **Centralized Identity & Access Management:** Creation and configuration of a Domain Controller (DC) to handle user authentication, account provisioning, and access permissions across the entire organization.
- **Organizational Structure (OUs):** Structuring the domain network into logical **Organizational Units (OUs)** based on company departments (e.g., HR, Finance, IT, Sales) to keep administrative tasks clean and organized.
- **Group Policy Management (GPO):** Deployment of domain-wide security policies, including password complexity rules, account lockout thresholds, automated software installations, and desktop environment restrictions.
- **Role-Based Security & Permissions:** Creation of Security Groups and Distribution Groups to grant folder permissions and resource access based strictly on an employee's job role.

- **Workstation & Asset Management:** Joining client computers and workstations to the domain network for centralized monitoring, remote control, and security compliance.
- **Network Hardening & Administrative Delegation:** Delegating specific management rights to junior IT support staff without sharing full administrative privileges, reducing security risks.

Chapter-3

Project Review

3.1 Feasibility Study

Before starting the implementation of the Active Directory Users and Computers (ADUC) project, a complete feasibility study was carried out to ensure the project can be successfully completed with available resources:

- **Technical Feasibility:** The project uses standard Microsoft technology (Windows Server OS, Active Directory Domain Services, and client virtual machines). These tools are well-supported, reliable, and easily executed using virtualization software like Oracle VirtualBox or VMware Workstation.
- **Operational Feasibility:** The system replaces complex manual user management with a simple centralized GUI interface (ADUC). It makes daily administration tasks (like resetting passwords or managing OUs) very straightforward for IT staff.
- **Economic Feasibility:** The project uses existing laboratory computers, free trial licenses of Windows Server, and free virtualization software. Therefore, the implementation cost is minimal.

3.2 Product Definition

The **Active Directory Users and Computers (ADUC)** system is a management snap-in provided by Microsoft Windows Server. It allows system administrators to centrally control domain identities, user credentials, client computers, organizational units (OUs), security groups, and group policies across a enterprise network.

3.3 Problem Statement

In traditional peer-to-peer networks without a centralized domain system:

1. IT staff must manually configure user accounts and passwords on every single computer individually.
2. There is no central tracking of user activities or file access permissions.

3. Security rules (like strong password policies and screen timeouts) cannot be enforced automatically across all computers.
4. Revoking access for a resigned employee is slow and error-prone, creating serious cyber security risks.

3.4 Project Functionality

The system provides the following main functionalities:

- **Centralized User Account Creation:** Easily create, modify, disable, or delete domain user accounts from one server console.
- **Organizational Structure (OU Management):** Group users, staff, and computers logically into OUs based on departments (e.g., HR, Finance, IT).
- **Security & Distribution Groups:** Assign users to security groups to grant or restrict access to shared folders and network resources.
- **Group Policy Object (GPO) Enforcement:** Automatically apply network-wide security rules (e.g., password length, software installation restrictions).
- **Administrative Delegation:** Delegate specific minor tasks (like resetting passwords) to junior IT support without sharing full admin rights.

3.5 System Requirements

Hardware Requirements (Server / Domain Controller):

- **Processor:** 1.4 GHz 64-bit Processor (Dual-core or Quad-core recommended)
- **RAM:** Minimum 4 GB (8 GB recommended for smooth VM performance)
- **Hard Disk Space:** Minimum 32 GB for OS installation and AD database (ntds.dit) storage
- **Network Interface Card (NIC):** 1 Gbps Ethernet Adapter

Software Requirements:

- **Operating System (Server):** Microsoft Windows Server 2019 / 2022 / 2025
- **Operating System (Client PC):** Windows 10 / 11 Enterprise or Pro edition

- **Virtualization Tools:** VMware Workstation or Oracle VM VirtualBox
- **Directory Services Role:** Active Directory Domain Services (AD DS) and Domain Name System (DNS)

3.6 Methodology

The project was executed following a structured step-by-step approach:

1. **Requirement Analysis:** Identifying the enterprise network layout, OUs required, and necessary security policies.
2. **Environment Setup:** Setting up Windows Server and Windows Client VMs on a virtual network.
3. **Domain Installation:** Installing the **Active Directory Domain Services (AD DS)** role and configuring a new Domain Controller (DC) along with DNS.
4. **OU & User Setup:** Designing the OU hierarchy and creating sample user accounts and security groups.
5. **Client Domain Join:** Joining client workstations (Windows 10/11) to the created domain network.
6. **GPO Configuration & Testing:** Applying Group Policies and verifying that rules and user permissions function correctly across client machines.

3.7 Acceptance Criteria

The project is considered complete and successful when:

1. The Domain Controller is actively running and properly serving DNS requests.
2. Client computers can successfully connect and join the created domain network.
3. Domain users can log into any connected client PC using their domain credentials.
4. Group Policies (GPOs) apply automatically upon user login (e.g., password enforcement, desktop restrictions).
5. Administrators can instantly update, disable, or modify user rights from the central ADUC console.

Chapter-4

Analysis

4.1 External interface and data flows(ER-diagrams)

In an Active Directory environment, the data flow revolves around central authentication, directory lookups, and policy distribution. Instead of a traditional database ER diagram, the structural architecture relies on a **Hierarchical Directory Tree**:

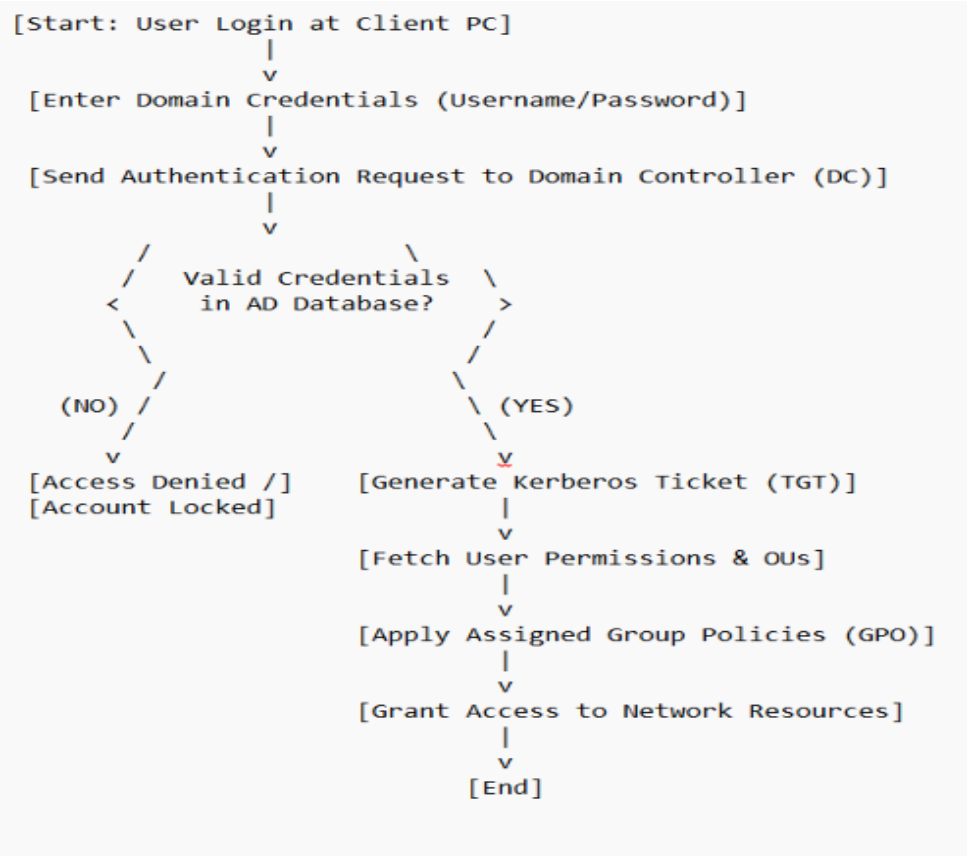


Fig:- 4.1

➢ Authentication & Interface Data Flow:

1. **User Login Request:** A client user inputs credentials (Username & Password) at a domain-joined workstation.
2. **Kerberos Authentication:** The request travels over the network to the Domain Controller (DC) running Active Directory Domain Services (AD DS).
3. **Identity Verification:** AD DS checks the central directory database (ntds.dit) and verifies the hash credentials and active status.
4. **Ticket & Policy Response:** Upon successful verification, the DC issues a Kerberos Ticket Granting Ticket (TGT) along with applicable Group Policy Objects (GPOs) to the user's workstation, granting secure access.

4.2 Flowcharts



A flowchart is a visual representation of a process, algorithm, or workflow, shown using standard geometric shapes connected by arrows. In software engineering and network administration, flowcharts break down complex operational logic into step-by-step sequential diagrams that are easy to understand.

4.3 User Display and Report Format

The primary user display interface used in this project is the Active Directory Users and Computers (ADUC) Management Console (dsa.msc).

- **Console Display**: A two-pane window where the left pane displays the domain tree, container folders, and Organizational Units (OUs), while the right pane displays detailed objects such as Users, Security Groups, and Computer Objects.

- **Object Properties Display:** Double-clicking any user or computer opens detailed tabbed displays (e.g., General, Member Of, Account, Profile, Environment).
- **System Reports & Audit Formats:**
    - **User Activity Logs**: Reports generated via Windows Event Viewer (Event ID 4624 for successful logins, Event ID 4625 for failed attempts).
    - **Group Membership Reports**: PowerShell exports (e.g., Get-ADD User, Get-ADD Group Member) listing all active users, locked accounts, or staff assigned to specific department OUs in .csv format.

4.4 Data Dictionary

- The data dictionary describes the primary attributes stored within the central Active Directory database (ntds.dit) for user and computer objects:

OBJECT TYPE	ATTRIBUTE NAME	DATA TYPE	DESCRIPTION / FUNCTION
User	`sAMAccountName`	String	Unique login ID for the user (e.g., `john_doe`)
User	`userPrincipalName`	String	Full domain login email style ID (e.g., `john@company.local`)
User	`userPassword`	Encrypted String	Stored password hash used to verify user login
User	`userAccountControl`	Bitmask Flags	Shows if the account is Enabled, Disabled, or Locked
User / Group	`objectSid`	Security Identifier	Unique code assigned by AD to identify every user or group
Group	`groupType`	Integer	Defines if the group is a Security or Distribution group
Computer	`dNSHostName`	String	Full network name of the connected computer
OU	`distinguishedName`	String	Full folder path showing where the user or computer is located

4.5 Functional and Performance Specifications

- **Functional Specifications:**
  - **Central Authentication:** Single Sign-On (SSO) functionality across all connected workstations and shared network resources.
  - **Granular Administration**: Ability to create, lock, disable, or reset passwords for any account instantly.
  - **GPO Deployment**: Automatic enforcement of desktop settings, password security rules, and software installations upon login.
  - **Role** Delegation: Ability to delegate simple admin rights (such as password resets) to specific helpdesk users.
- **Performance Specifications:**
  - **Authentication Speed**: User authentication and Kerberos ticket issuance completed in under 2 seconds over a standard Local Area Network (LAN).
  - **GPO Processing Time**: Group Policy updates applied within 3 to 5 seconds during system startup/user login.
  - **System Availability:** High domain service uptime ensured through Domain Controller backup and DNS integration.
  - **Database Efficiency:** Lightweight directory access protocols (LDAP) allow high-throughput querying of thousands of objects simultaneously.

Chapter-5

Project Plan

5.1 Development Schedule

➢ Overview

This chapter outlines the project management strategy, execution timeline, and the technical environment utilized during implementation. A structured project plan ensures the efficient allocation of time and resources to meet all functional requirements.

➢ Development Schedule

The project implementation was completed over a period of 1 month (4 weeks). The schedule was structured sequentially to ensure proper infrastructure setup, domain configuration, security policy deployment, and system verification.

```
Phase 1: Planning & Requirements
 ---> Phase 2: Design & Architecture
       ---> Phase 3: Core Module Implementation
             ---> Phase 4: Integration & Testing
                   ---> Phase 5: Deployment & Documentation
```

➢ Weekly Project Time

Week	Project Phase	Key Tasks & Deliverables	Status
Week 1	Virtual Infrastructure Setup	• Installed Oracle VirtualBox software. • Created virtual machines for Windows Server 2022 and Windows 10. • Configured CPU, memory, and disk settings.	Completed
Week 2	Network & Domain Setup	• Assigned static IP address (192.168.1.241) and DNS parameters. • Installed Active Directory Domain Services (AD DS). • Configured local domain (AES256.in).	Completed
Week 3	AD Architecture & GPO Deployment	• Created IT and Sales Organizational Units (OUs). • Provisioned domain accounts (Member 1). • Configured Group Policy: "Prohibit access to Control Panel and PC settings".	Completed
Week 4	Testing & Documentation	• Applied policies using gpupdate /force. • Verified restrictions on Windows 10 target machine. • Collected screenshots and completed report documentation.	Completed

5.2 Programming Language and Development Tools

The implementation of the domain-controlled network environment and policy management framework relies on a combination of enterprise operating systems, virtualization software, system administration utilities, and management protocols rather than traditional software programming languages.

➢ Operating Systems & Virtualization Platform

Windows Server 2022 (Datacenter / Standard Edition): Serves as the primary server operating system hosting Active Directory Domain Services (AD DS), DNS, and Group Policy Management infrastructure.

Windows 10 Enterprise / Pro: Acts as the client workstation joining the AES256.in domain to test

policy enforcement and user restriction rules.

Oracle VM VirtualBox (v7.0+): Provides the Type-2 hypervisor environment to host and interconnect the server and client virtual machines on isolated virtual networks.

➢ Server Management Tools & Administration Utilities

Active Directory Users and Computers (ADUC): Used for administrative structuring, including creation of Organizational Units (OUs) such as IT and Sales, user provisioning (Member 1), and group privilege assignment.

Group Policy Management Console (GPMC): The centralized management interface used to construct, edit, link, and enforce Group Policy Objects (GPOs) across the domain tree.

Server Manager Dashboard: Provides centralized monitoring and management of server roles, services, network configurations, and storage resources.

➢ Command Line Interfaces & Administrative Tools

Windows PowerShell / Command Prompt (cmd): Executed administrative commands for domain auditing, policy refresh routines (gpupdate /force), network diagnostics (ipconfig /all, ping), and credential verification .

Remote Desktop Connection (RDP): Standard protocol utility used for secure remote session administration of the domain controller machine across the private subnet.

Chapter-6

Design

In an Active Directory Domain Services (AD DS) network environment, design focuses on operational data workflows, administrative access pathways, user provisioning, policy distribution, and the directory object schema.

6.1 Data Flow and Operational Workflow

The operational workflow inside a domain-controlled architecture consists of server initialization, remote session setup, user/OU object creation, Group Policy Object (GPO) configuration, policy enforcement update, and restriction verification.

➢ Server Network Adapter Configuration Flow

To establish a reliable domain controller for AES256.in, static network parameters are assigned to the network interface card (NIC).

- **Workflow:** Network Adapter Properties → IPv4 Settings Selection → Static IP & DNS Binding.

➢ Static Network Addressing Data Flow

Static IPv4 routing and loopback DNS resolution are assigned to ensure proper domain name resolution across client workstations.

Workflow: IP Address: 192.168.1.241 → Subnet Mask: 255.255.255.0 → Preferred DNS: 192.168.1.241.

➢ Remote Desktop Protocol (RDP) Session Setup Flow

Centralized administrative control is established using Remote Desktop Protocol to manage the Windows Server 2022 instance over network port 3389.

- **Workflow:** Remote Desktop App → Target IP 192.168.1.241 → Enterprise Admin User Authentication (EKAMPREET\Administrator).

6.2 Data base tables

1. User Accounts Table (User Object Class)

Field/Attribute	Data Type	Example Value	Description
sAMAccountname	String	Member 1	Pre-windows 2000 user logon name
userPrincipleName	String	member1@AES256.in	Full UPN format logon identifier
distinguishedName	String	CN=Member 1,OU=Sales,OU=AES256,DC=AES256,DC=in	Complete X.500 path in Active Directory
userAccountControl	Integer	512	Bitmask flag for account
memberOf	Array	CN=Administrators,CN=Builtin,DC=AES256,DC=in	Group memberships assigned to the account

2. Organizational Units Table (organizationalUnit Object Class)

Field/Attribute	Data Type	Example Value	Description
ou	String	Sales	Name of the Organizational Unit container
distinguishedName	String	OU=Sales,OU =AES256, DC=AES256, DC=in	Hierarchical directory path of the OU
gPLink	String	[LDAP://cn={GUID}...;0]	Link binding Group Policies to this OU
objectGUID	GUID	a1b2c3d4-e5f6-...	128-bit unique internal

Chapter-7

Results and Discussion

7.1 Snapshots with Description

This chapter details the implementation results and functional testing of the Active Directory domain infrastructure for AES256.in.

➢ Group Policy Restriction Enforcement

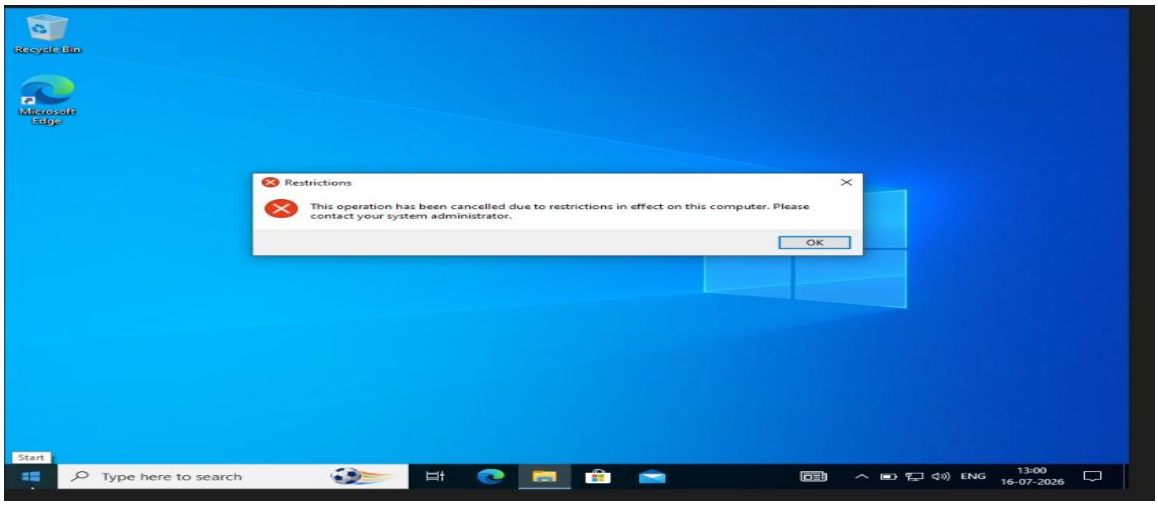


Fig7.1.1

- **Description**: Shows the Windows 10 restriction dialog blocking Control Panel access for Member 1.
- **Discussion**: Confirms successful GPO application. Standard domain users are prevented from changing OS and network settings.

➢ Remote Desktop Configuration

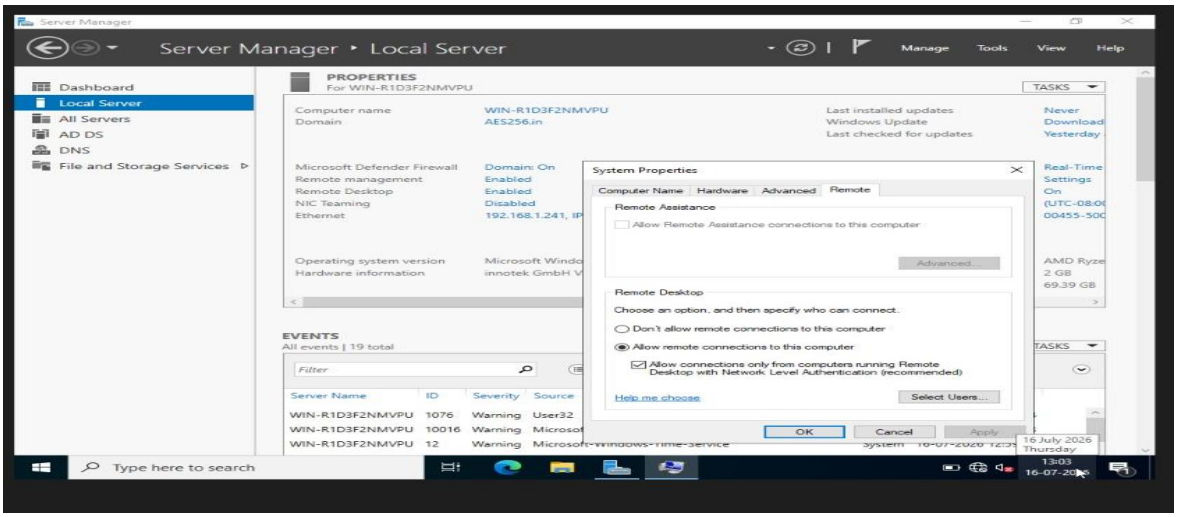


Fig7.1.2

- **Description**: Displays System Properties on Windows Server 2022 with Remote Desktop and NLA enabled.
- **Discussion**: Enables secure remote server administration over port 3389 while restricting unauthorized access.

➢ Active Directory Domain Root

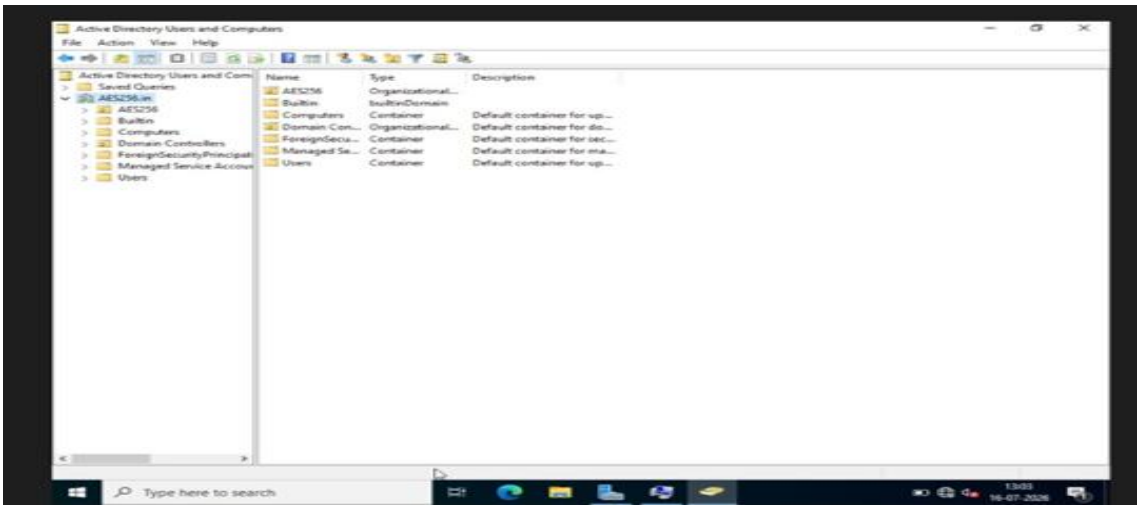


Fig7.1.3

- **Description:** Displays the ADUC console showing the AES256.in domain root and top-level AES256 OU.
- **Discussion**: Verifies active domain installation and structured directory hierarchy.

➢ Departmental Organizational Units

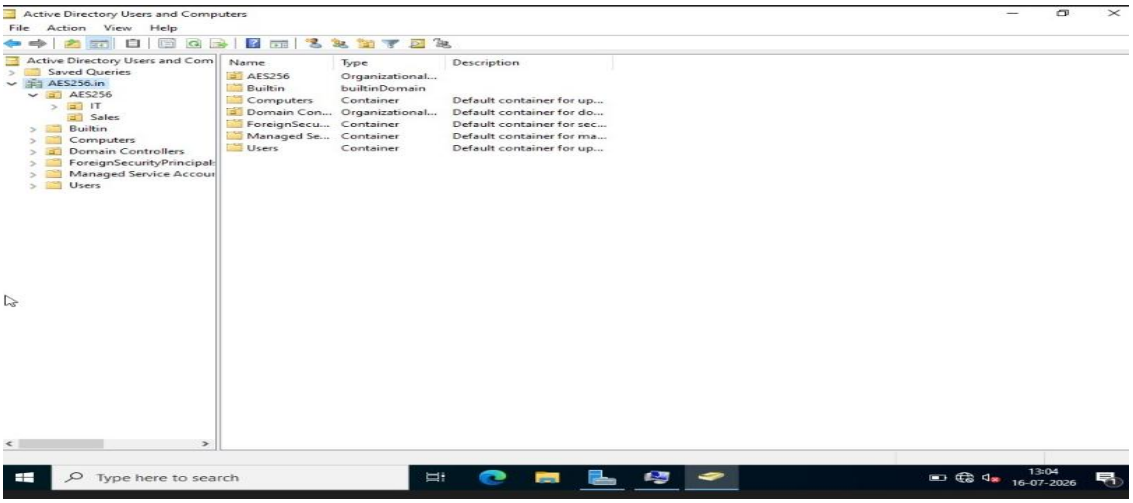


Fig7.1.4

- **Description:** Shows sub-OUs IT and Sales created inside the AES256 container.
- **Discussion**: Establishes administrative boundaries to target policy settings to specific departments.

➢ User Account Provisioning

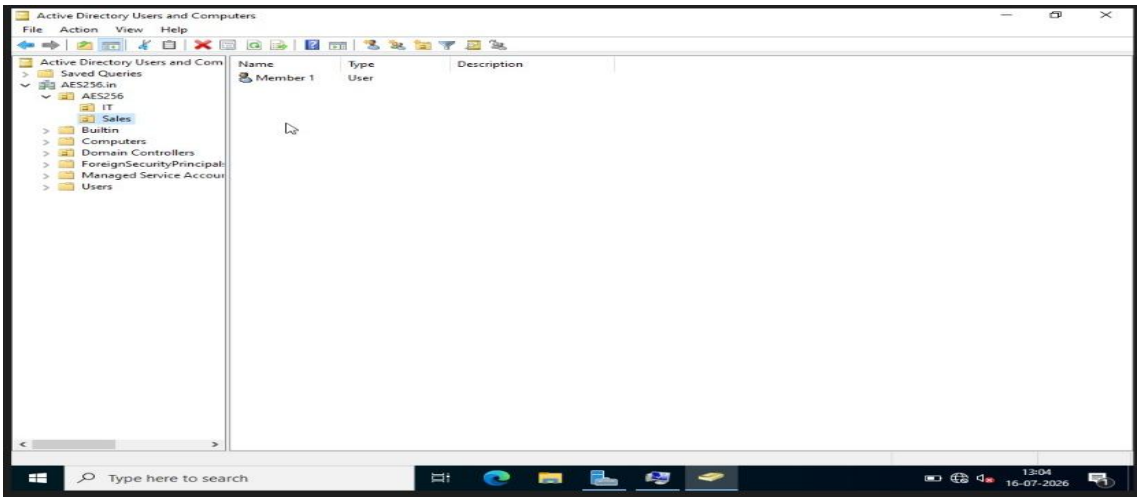


Fig7.1.5

- **Description:** Displays the Member 1 user account created inside the Sales OU.
- **Discussion:** Confirms user object creation within its designated container for proper policy inheritance**.**

Chapter-8

CONCLUSION AND FUTURE SCOPE

This chapter summarizes the administrative outcomes of the Active Directory Domain Services (AD DS) deployment and outlines future technical enhancements for expanding domain management, security enforcement, and infrastructure scalability.

8.1 Conclusion

The deployment of Active Directory Domain Services (**AD DS**) for the **AES256.in** domain was successfully implemented and verified across both server and workstation environments. Key achievements of this project include:

1. Centralized Identity Management: Established a domain directory hierarchy (**AES256.in)** with top-level and departmental Organizational Units (**AES256, IT, Sales**), enabling structured domain account provisioning and role-based access control (RBAC).
2. Static Network & DNS Infrastructure: Configured static network binding (**192.168.1.241**) and local DNS hosting on Windows Server 2022 to guarantee uninterrupted domain lookup, Kerberos authentication, and remote management access via Remote Desktop Protocol (RDP).
3. Targeted Group Policy Enforcement: Created and linked custom Group Policy Objects (**control panel blocked**) to enforce system restrictions on client workstations, successfully blocking unauthorized non-administrative access to system settings and network configurations.
4. Verified Policy Propagation: Validated real-time GPO synchronization and client update execution using **gpupdate /force,** confirming proper registry enforcement and active communication between the client endpoints and server **SYSVOL** shares.

8.2 Future Scope

As a 5th-semester B.Tech student, this project gives a strong foundation in core Computer Networks and System Administration. To make this Active Directory setup more advanced and industry-ready, the following future enhancements can be added:

1. Adding a Backup Server (Domain Redundancy)
    - What to do: Set up a second Domain Controller (Backup DC).
    - Why: If the main server fails or goes offline, the backup server will instantly take over so users can still log in without any network downtime.
2. Connecting to Cloud (Hybrid Active Directory)
    - What to do: Connect the local Windows Server to cloud services like Microsoft Azure AD (Entra ID) using Azure AD Connect.
    - Why: This allows users to use a single username and password to log in to both local college/office PCs and cloud apps (like Microsoft 365 or Google Workspace).
3. Adding Multi-Factor Authentication (MFA)
    - What to do: Enable OTPs or mobile app approval (like Microsoft Authenticator) for administrative logins.
    - Why: Passwords alone can be guessed or hacked. Adding a second verification step makes remote desktop (RDP) and server logins much more secure.
4. Automating Admin Tasks with PowerShell
    - What to do: Write custom PowerShell scripts to create hundreds of student/employee accounts from an Excel sheet automatically.
    - Why: Manually creating users one by one in Active Directory takes too much time. Scripting helps automate daily IT management tasks efficiently.
5. Implementing Network Access & Certificate Security (AD CS)
    - What to do: Install Active Directory Certificate Services (AD CS).

- Why: It allows the domain to issue digital security certificates for Wi-Fi authentication (802.1X) and secure device identification on the network.

Chapter-9

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